

FFAR Final Progress Report

As part of closing out a FFAR grant, grantees must complete a Final Progress Report. Please use the template below to complete the programmatic report. The report must be submitted to FFAR within 90 days after the expiration or termination of a FFAR funded research grant. All questions about this form should be directed to grants@foundationfar.org.

The Final Progress Report communicates the cumulative results and major accomplishments of the funded grant research, including accomplishment for each aim and whether all research aims were fully completed. FFAR uses the content of the Final Progress Report to inform Congress, the Advisory Board, and other stakeholders on the successes, impacts and value of funding unique partnerships to support innovative science addressing today's food and agriculture challenges. If a question is not applicable to the project, please type "Nothing to report."

Grant Information	
Grant ID	593569
Award Program	New Innovator Award
Project Title	Optimizing Water Use in Agriculture by Stacking Conservation Practices
Grant Start Date	12/01/2019
Grant End Date	06/30/2022
Total Grant Budget	\$600,000
Total FFAR Budget	\$300,000
Matching Funder(s)	Utah Agricultural Experiment Station, Utah State University Office of Research and Graduate Studies, Central Utah Water Conservancy District, Uintah and Iron/Enterprise County Conservation Districts
Final Report Date	09/01/2022
Project Director/Principal Investigator Information	
Full Name	Matt Yost
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Grantee Organization Information	
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Authorized Signing Official (ASO) Information	
Name	Devin Hansen
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General Information

- 1.1. Please list the geographic location(s) – city, state, congressional district - where the work was conducted. If the work was conducted outside of the US, please list the city and country.
Logan, Cedar City, and Vernal, UT
- 1.2. How many new jobs were created because of this grant funding?
We hired two graduate students and five undergraduate students.
- 1.3. How many jobs were maintained because of this grant funding?
All jobs associated with accomplishing the work were temporary student jobs.
- 1.4. Have there been any changes to your organization's IRS 501(c)(3) non-profit status since you were awarded the grant? If yes, please explain.
No
- 1.5. Has your organization undergone a recent name change? If so, please provide the new name of your organization.
No

2. Scientific Report

- 2.1. Public Abstract (up to 500 words): The abstract should be 'stand-alone' and is intended for a general audience. It should be a concise overview/summary of the importance of the project, the issues that the project addressed, and the key findings of the project. The abstract should also clearly state how the results of the project help to address important needs in U.S. food and agriculture systems.

This project sought to provide agricultural producers and water managers with tools for optimizing agriculture water use. Utah State University partnered with Southern Utah University, the irrigation industry, Water Conservancy Districts, soil and water conservation districts, Utah water agencies, and several other federal



and state organizations to evaluate and demonstrate over 25 different water optimization practices. The major objective was to: identify which combinations of pivot irrigation and crop management practices result in optimized use of limited water supplies, reduced consumptive use, and the best yield and profit outcomes for producers.” The trials include evaluations of:

- Pivot irrigation technologies such as mobile drip irrigation (MDI), low-energy precision application (LEPA), and low-elevation spray application (LESA),
- Deficit irrigation strategies,
- The best available drought-tolerant crop genetics,
- Reduced and no-tillage practices,
- Cover crops, and
- Alternative crops with lower water requirements.

These side-by-side evaluations are the first of their kind and were established in Logan in 2019, Vernal in 2020, and Cedar City in 2021. This information has and will be especially useful in guiding water conservation planning at the farm level, which would in turn have large impacts on planning efforts at watershed and basin levels. It will also help irrigators prepare to effectively participate in water demand and banking programs, should they be developed and necessary. Results showed that irrigation rate has the most potential of factors tested to improve water optimization. This was followed closely by irrigation technologies. Crop genetics and targeted deficit irrigation rarely improved crop production in water stress conditions. Further, no-tillage and cover crops also did improve the ability of corn to withstand water stress in the short term. This study has become the basis of a long-term water optimization effort that will continue to identify ideal ways to optimize water use in agriculture with a combination of water, crop, and soil management.

- 2.2. Provide a description or interpretation of how the key findings of the project can be used by the agriculture community. It is important the PI identifies how these findings have been, or may be, adopted or adapted in U.S. agriculture and food systems. If the findings cannot yet be applied, this section should address how they can be used to guide future planning or decision-making (up to 1,000-word limit).

The information from this study about water optimization practices has and will be especially useful in guiding water conservation planning at the farm level, which would in turn have large impacts on planning efforts at watershed and basin levels. It will also help irrigators prepare to effectively participate in water demand and banking programs, should they be developed and necessary. One example is an irrigation investment calculator that we built (in part) using data collected from this study. This [tool](#) guides investments in water optimization practices.

- 2.3. Research Outcomes/Impact (up to 1,500-word limit). The primary goal here is

to answer questions such as “How did this project lead to improvements in U.S. agriculture and food systems?” or “How can the findings of this study guide or make improvement in future investigations and research?” Outcomes should be explained and classified in one of the following ways:

- a) potential outcomes, i.e., findings, results, or recommendations that could impact U.S. agriculture and food systems, if used;

Results showed that irrigation rates had the largest impact on water optimization. This was followed closely by irrigation technologies, which had site-specific benefits to water optimization and crop production. Crop genetics and targeted deficit irrigation rarely improved crop production in water stress conditions. Further, no-tillage and cover crops also did improve the ability of corn to withstand water stress in the short term. This study has become the basis of a long-term water optimization effort that will continue to identify ideal ways to optimize water use in agriculture with a combination of water, crop, and soil management.

- b) intermediate outcomes, i.e., how the findings, results, or recommendations have been used by others to influence practices, legislation, research/product design, training and so forth; and

This study has become the basis of a long-term water optimization effort that will continue to identify ideal ways to optimize water use in agriculture with a combination of water, crop, and soil management. These efforts have guided legislation in Utah to provide \$70M to incentive water optimization on Utah farms, and has guided priorities of the Utah Water Optimization Task Force who received \$2M to further advance water optimization in the state of Utah. Further, it has become the basis of other demand management program efforts by water conservancy districts in Utah to invest millions of dollars each year to identify how to best work with agriculture to bank water to meet needs of the Great Salt Lake and the Upper Colorado River Basin.

- c) end outcomes, i.e., how findings, results, or recommendations have contributed to documented reductions in work-related morbidity, mortality, and/or exposure.

This study will continue to be the basis of a long-term water optimization effort that identify ideal ways to optimize water use in agriculture with a combination of water, crop, and soil management. They will continue to guide future policy and investments in irrigation technology improvements,



water right laws, and other investments in water optimization by Utah governments, organizations, and other institutions. It has and will continue to influence other states in the Western U.S. facing similar devastating droughts and water issues.

- 2.4. What were the goals/specific aims of the project? If the approved application lists milestones/target dates for important activities or phases for this reporting period, identify these milestones and dates, as well as show actual completion dates or the percentage of completion of milestone targets. (up to 1,500-word limit)

The goals and aims of this project were to establish the three long-term irrigation research fields in Utah where multiple water conservation practices can be investigated; compare the water use efficiency, productivity, and economics of various combinations of water conservation practices to identify the most ideal combinations; educate stakeholders by holding a field days to share results; and educate and train graduate students. These aims were all accomplished on our proposed timeline. All three sites are fully established and we have published over 15 extension articles on results from the first three growing seasons of data. Results have also been presented to over 1,000 stakeholders in various formats in Utah, the west, and in international conferences. Two graduate students were trained and have or will shortly complete their Master's thesis.

- 2.5. Have any of the major goals/specific aims or milestones for the project changed since the award? If so, please list the goal(s) that have changed and provide justification for the change from the approved goals. (up to 500-word limit)

There were no major changes.

- 2.6. What was accomplished under the goals/specific aims or milestones for the project? Please describe in detail, 1) major activities; 2) specific objectives; 3) significant results, including major findings, developments, or conclusions (both positive and negative); and 4) key outcomes or other achievements. Include discussion of stated goals not met. The emphasis in reporting in this section should focus on accomplishments. In the response, emphasize the significance of the findings to the scientific field. This section can be as technical as the PI would like. (up to 5,000-word limit)



1) Major activities:

The first major activity was to establish three long-term irrigation sites near Logan, Vernal, and Cedar City Utah and to conduct three seasons (2019, 2020, and 2021) of research at the sites. This was accomplished. It included installing a lateral irrigation system on 20 acres land, and establishing nearly 400-700 research plots that included irrigation, crop, and crop management treatments at each site. A second major activity was to install a large soil moisture monitoring network at the site. This included successful installation of over 300 soil moisture content sensors. A third major activity was to hire and train graduate students. Tina Sullivan was recruited at the end of 2019 to start her M.S. degree. We also recruited a second graduate student – Dakota Boren to help continue the work in 2021. A fourth major activity was hosting field day, training events, and various outreach and education activities. We accomplished this with many field days, tours, training events, and presentations of results at over 20 extension and research events. A fifth major activity that was accomplished including collecting yield and water use data from each research plot and publishing the data in peer-reviewed manuscripts and extension articles and outlets.

2) Specific objectives:

The objectives were to: 1) Establish the three long-term irrigation research fields in Utah where multiple water conservation practices can be investigated; and 2) Compare the water use efficiency, productivity, and economics of various combinations of water conservation practices to identify the most ideal combinations; 3) Educate stakeholders by holding a field days to share results; and 4) Educate and train graduate students.

3) Significant Results:

Corn. Irrigation technologies, irrigation rates, and drought-tolerant genetics had only minor impacts on yield in 2019 at the Logan site due to an abnormally wet spring. In contrast, most factors influenced silage corn yield in 2020 in Logan, and at the new established site in Vernal. The interaction of irrigation technology and irrigation rate was significant at both sites, with results suggesting that the higher efficiency sprinklers (LEPA, and LESA) can often result in higher yields when irrigation rates are reduced. Treatment type was heavily influenced by the irrigation technology as well at both sites. At reduced irrigation rates, drought-tolerant genetics never improved yield. No-till only sometimes decreased yield and reductions were more pronounced when water stress was greater. Targeted 50% irrigation rate reductions to critical crop growth stages was often worse and never better than a straight 50% reduction in rate all season.

Alfalfa. Two cuttings of alfalfa were harvested in 2020 in Logan. A total of 13.5 inches of irrigation was applied for both cuttings for the full irrigation rate. The irrigation reductions were achieved by changing nozzle sizes on the linear where alfalfa received 25 or 50% less water each irrigation. When irrigation was reduced by 25%, alfalfa yield was reduced by an average of about 0.75 tons/acre (20%) total for the two cuts. A 50% reduction in irrigation produced 1.2 tons/acre less



yield (30% reduction). Results were mixed for the 50% targeted irrigation. It was much worse than the straight 50% reduction for the first cut, but much better in the second cut.

Small Grains Forage. Small grain forage yield in 2020 in Vernal ranged from 1.5 to over 3.5 tons per acre. The LESA system at full irrigation produced the greatest yield, while the MDI produced the least. All systems except LESA were able to maintain yield with 25% less water. MESA was the only system where the 50% reduction also maintained yield compared to full irrigation. This indicates that small grain forage has a great ability to withstand water stress, and that traditional MESA systems may have some of the greatest potential. Small grain forage quality was only impacted by irrigation rates. Forage quality increased as water stress increased. This is likely because less growth in water-stressed areas can reduce plant stem, which can improve forage quality. Quality and yield data will be combined to examine the economics of these treatments.

Alternative Crops. Teff was the alternative crop in Vernal. At full irrigation, all irrigation systems besides MDI (poor yield in all treatments) performed equally well. None of the three advanced irrigation systems could maintain yield with 25% less irrigation. However, for MESA and LESA, the 50% reduction yielded the same as the 25% reduction in irrigation rate.

4) Key Outcomes:

Key outcomes included collection of data on how 16 irrigation and 5 crop/soil treatments influenced yield and water use of corn, alfalfa, small grains, and alternative crops (safflower, teff, hemp). Hosting a virtual and face-to-face field days with over 300 participants. Disseminating results in several fact sheets, press articles, and presentations at local, regional, and international meetings. Training of several graduate and undergraduate students. Securing additional grants to support expansions of this research.

2.7. Discuss efforts taken to ensure the approach is scientifically rigorous.

Include approaches taken to ensure robust and unbiased results.

We followed all standard procedures for agricultural research including replication and randomization of treatments, and careful documentation of all measurements and procedures.

2.8. List key stakeholders that could be served by results of this project.

Farmers, water managers, crop advisors, agricultural companies, policy-makers, conservation planners, and related stakeholders.

2.9. How have the results of this reporting period been disseminated to communities



of interest? Describe how the results have been disseminated to communities of interest. Include any outreach activities that have been undertaken to reach members of communities who are not usually aware of such activities, to enhance public understanding and increase interest in learning and careers in science, technology, and the humanities. Reporting the routine dissemination of information (e.g., websites, press releases) is not required. For awards not designed to disseminate information to the public or conduct similar outreach activities, a response is not required; the grantee should write "nothing to report." A detailed response is only required for awards or award components that are designed to disseminate information to the public or conduct similar outreach activities. Note that scientific publications and sharing research sources will be reported under Information Products. (up to 1,000-word limit)

The results of this project were disseminated in many ways through various Extension channels. We posted information and results on our social media accounts (USUcrops) that have nearly 1,500 followers. We also shared the results on our USUcrop website that gets several thousand visits each year.

We hosted six virtual events to disseminate the results due to CO-VID. A virtual field day in 2020 with over 160 participants to highlight the results. We also hosted virtual crop schools in 2021 where results were shared with farmers and other stakeholders. There was a total of 1,050 attendees at the five virtual crop schools! The first crop school was on alfalfa and it had a total of 300 people attend. This was the largest Zoom meeting that we have ever hosted. Most of the other sessions had about 200 attendees each. The crop schools were held in the evening and included presentations and panel discussions followed by several question-and-answer sessions. Further, we recorded all the presentations and made them available as individual YouTube videos. There have been over 3,000 views of the crop school presentations in the last eight months. Pesticide use credits (required for pesticide applicators) were offered in four of the five sessions. We issued a total of 495 credits total for the four events. Attendees joined from almost every county in Utah, along with some from neighboring states. Nearly 600 attendees completed the meeting evaluations. Below are some highlights of the evaluations:

- 98% rated their overall experience as "good", "very good", or "excellent"
- 93% indicated that they would recommend the event(s) to others
- Most farmers who attended had more than 30 years of experience (41%) but we also reached many farmers with 1-30 years of experience
- Attendees who completed evaluation represented 257,000 acres and advised farmers on over a million acres of farmland

We also held irrigation dealer trainings to demonstrate the technologies and educate on best water optimization practices. In 2019, we worked with an industry partner, WISH NW, INC (irrigation distributor for most of Western United States)



to co-sponsor and invite all irrigation dealers (~35) in Utah and Southeastern Idaho to attend a 2-day irrigation workshop in Wellsville, Utah at our water optimization trial. This was the first time this industry has come together for a workshop, and it was a large success. There were 50 attendees, including nearly all the irrigation dealers in Utah. These dealers service irrigation systems on over 2.5 million acres of farmland. The workshop included reports of recent irrigation-related research, industry presentations, demonstrations and tours of irrigation research trials, and discussions on irrigation research and educational needs. All irrigation dealers who completed evaluations indicated that the workshop should be repeated and that they gained knowledge on best irrigation practices. Eight SU attendees also attended and had significant knowledge gains.

Results of the study were also presented at various in-person crop schools and field days where we educated farmers, water managers, state agency personnel, and various other stakeholders. In addition to presenting results at various events, we also disseminating results in several fact sheets, press articles, and presentations at regional, and international conferences.

- 2.10. Describe challenges or delays encountered during project and actions that weretaken to resolve them. Only describe significant challenges that may have impeded the research and emphasize their resolution. (up to 500-word limit)

One challenge we had was establishing a functioning remote access soil moisture sensing network. The company that we purchased sensor networks from (Acclima) ensured us that their telemetry was operational, however, it was not and there were several updates and other issues that had to be resolved. These were not resolved in 2020 and we lost nearly all the soil moisture data for the Logan site. The company has committed to fixing this problem for the 2021 season and has sent a whole new set of dataloggers for 2021. We switched to another company for the Vernal and Cedar City sites and they have worked extremely well so far.

- 2.11. What opportunities for training and professional development has the project provided during this reporting period? If the research is not intended to provide training and professional development during this period, state "Nothing to Report." For all projects reporting graduate student and/or post-doctoral participants, grantees are encouraged to describe how Individual Development Plans (IDPs) are used to help manage the training for those individuals. (up to 500-word limit)

This grant allowed for the hire of two master's level graduate student. These students have received training in proper research methods, learned to operate new machinery, program dataloggers, communicate results to stakeholders at field day, manage undergraduate students, operate irrigation systems, and manage



farming operations on 40 acres. The graduate students were also fully engaged in our Department professional development course, along with regular course work requisite for graduate students.

- 2.12. Please indicate the number of undergraduate and graduate students, post-doctoral scholars, or other educational components involved during this reporting period. If other education components are involved, please describe them in detail. (up to 500-word limit)

We hired two graduate students on this project and five undergraduate students. This has resulted in one master's thesis and will result in another thesis in December 2022.

3. Information Products

- 3.1. Please list the type(s) of information products (e.g., scholarly publications, reports or monographs, workshop summaries or conference proceedings, video, audio, images, models, software, curricula, instruments or equipment, intervention, etc.) produced during the project resulting directly from the FFAR award.

We produced numerous extension fact sheets, guides, farm press articles, conference and extension presentations.

- 3.2. Please provide a list of citations for the information products produced during the project.

Fact Sheets

<https://extension.usu.edu/crops/research/4rs-of-irrigation-management>

<https://extension.usu.edu/crops/research/defense-against-drought>

<https://extension.usu.edu/crops/research/irrigation-pivots-laterals>

<https://extension.usu.edu/crops/research/irrigation-water-loss-and-recovery>

<https://extension.usu.edu/crops/research/mobile-drip-irrigation-for-pivots-and-laterals>

<https://extension.usu.edu/crops/research/strategies-for-deficit-irrigation-of-forage-crops>

<https://extension.usu.edu/crops/research/variable-frequency-drives-for-irrigation-pumps>

<https://extension.usu.edu/crops/research/measuring-and-building-soil-health>

<https://extension.usu.edu/crops/research/ten-frequently-asked-questions-about-growing-cannabis>

<https://extension.usu.edu/irrigation/research/25-rules-of-thumb-for-field-crops>



<https://extension.usu.edu/crops/research/ten-reasons-why-alfalfa-is-highly-suitable-for-the-west>

Articles

1. Irrigation investment tool. <https://mailchi.mp/0bde7b01d265/irrigation-tool>.
2. Drought defense tips: Deficit irrigation. <https://mailchi.mp/3054954d53ef/droughtdefense>.
3. Drought defense tips: Crop diversification. <https://mailchi.mp/8ea1533910f7/droughtdefense>.
4. Drought defense tips: Pivot sprinkler innovations <https://mailchi.mp/86efda446e22/droughtdefense>.
5. Advanced pivot irrigation systems help save water. extension.usu.edu/news_sections/impacts/agriculture.
6. Agriculture water optimization. research.usu.edu/landscapes/wp-content/uploads/sites/64/2019/06/Yost.pdf.

Videos

1. Soil health sampling. <https://www.youtube.com/watch?v=5gSYIuTIvDs&t=5s>.
2. Hemp field research. <https://youtu.be/5OZzBJctB2E>.
3. Corn water optimization studies. <https://youtu.be/gVMQKBy9gCc>.
4. Alfalfa irrigation updates. <https://youtu.be/j7bOsRaa1tM>.
5. How does hemp establish under pivot irrigation? <https://youtu.be/I2ledRto7v0>.
6. Stacking conservation practices <https://youtu.be/uJEGUnBUXw0>.

- 3.3. Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication and monograph) during the project resulting directly from the FFAR award? If yes, please provide citation.

Yost, M., N. Allen, W. Peterson, and J. Gale. 2020. Water conservation through applied irrigation research. 75 Year of Conservation Special Edition. *Journal of Soil Water Conservation*. https://www.swcs.org/static/media/cms/75th_Book_online_small_EB8351774A8B2.pdf.

Sullivan, T., M. Yost, D. Boren, B. Kitchen, E. Creech, R. Violett, and B. Barker. (2022). Impacts of irrigation technology, irrigation rate, and drought tolerant genetics on silage corn production. *Agronomy Journal*. In review.

Sullivan, T., M. Yost, E. Creech, N. Hansen, B. Hopkins, O. Walsh, and J. Williams. (2022). Irrigation and nitrogen 4R opportunities in the Western United States. *Agronomy Journal*. Will be submitted by 9/30/2022.

- 3.4. Website(s). List the URL for any internet site(s) that disseminates the research activities. A short description of each site should be provided. It is not necessary to include the publications already specified above.

<https://extension.usu.edu/crops/>. This website was developed to disseminate extension materials associated with this and several other projects.

- 3.5. Have inventions, patent applications, and/or licenses resulted from the project (U.S. and international)? If yes, indicate the invention, patent application(s), and/or license(s).

No

	# Filed (Enter Numeric Value)	# Approved (Enter Numeric Value)	Patent Numbers (Separated by commas)
Patents	#0	#0	0
Copyrights	#0	#0	
Trademarks	#0	#0	
Inventions	#0	#0	

- 3.6. Are any of the information products produced during this grant that are confidential, proprietary, or subject to special license agreements? If so, please list them below and describe why they must remain confidential. Also, note if (and when) you plan to make these data publicly available in the future or if they must remain confidential indefinitely. (up to 500-word limit)

No

- 3.7. Beyond depositing information products in a repository, what other activities have you undertaken to ensure that others (e.g., researchers, decision makers, and the public) can easily discover and access the listed information products? What other activities have you undertaken to ensure that others can access and re-use these data in the future?



We have published much of the data in fact sheets, presentations, and publications. As manuscripts are published, we will provide open access to data associated with each manuscript in supplemental materials.

4. Data Management

- 4.1. Did the project generate any data? Data generation includes transformation of existing data sets and data from existing resources (e.g., maps and imagery). Please list the data generated for the award.

Yes. Soil moisture, soil nutrition, plant nutrition, forage quality parameters, crop yield, water inputs.

- 4.2. If you list multiple data sets, are these data sets related? If so, please provide a short description of how they are related.

The data sets are all soil, plant, and water data that are all highly related.

- 4.3. Please provide copies of relevant metadata records to support FFAR's mission of enhancing the discoverability of FFAR funded project data and information products. Include or attach copies of records and a simple file inventory, if necessary, in a compressed folder.

We currently don't have metadata for these data but will publish in manuscripts as they are completed.



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RESEARCH

Certification

The undersigned hereby certifies that to the best of their knowledge and belief, the above report and all supporting documents are true, accurate and complete. I am aware there is a significant penalty for submitting false or misleading information.

If applicable, I agree that my electronic signature is legally binding, equivalent to, and has the same validity and meaning as my handwritten signature. I will not claim otherwise.

PI Full Name: Matt Yost

PI Signature:  Date: 09/01/2022

Authorized Signing Official (ASO) Name: Devin Hansen

ASO Signature:  Date: 09/01/2022